

Detailed Project Decisions and Recommendations

Dataset Selection Decision

The historical sales transaction dataset was selected because it contains detailed information related to customer purchases, product information, market details, and operational variables. Sales forecasting depends heavily on historical behavior because previous sales patterns often provide important signals for future demand estimation.

The selected dataset includes variables such as quantity, discount, shipping cost, category, market, region, and sales revenue. These variables were considered useful because they influence customer purchasing behavior and business performance. Historical datasets also help machine learning algorithms identify trends, seasonal patterns, and hidden relationships among variables.

Recommendation:

Organizations should collect large and high-quality historical sales data because prediction accuracy depends heavily on data quality and availability.

Missing Value Handling Decision

Missing values are common in real-world datasets and may negatively affect model performance if not handled correctly. During data exploration, missing values were identified within the product name feature.

Since the missing observations represented only a small portion of the dataset, removing rows would unnecessarily reduce available information. Therefore, mode imputation was selected because it replaces missing values with the most frequently occurring category.

This approach maintains dataset size while preserving data consistency.

Recommendation:

Organizations should establish better data collection systems to minimize missing information because poor data quality can reduce prediction accuracy.

Duplicate Record Decision

Duplicate records were checked because repeated transactions can introduce bias into machine learning models. Duplicate observations may cause the model to overlearn certain patterns and generate misleading predictions.

Duplicate entries were removed to ensure each transaction contributed equally during training.

Recommendation:

Companies should periodically monitor databases for duplicate records because duplicated information can distort business analysis.

Outlier Detection and Handling Decision

Sales values contained extreme observations significantly larger than normal values. Outliers may arise because of data entry errors, exceptional events, promotional campaigns, or seasonal peaks.

Initially, the Interquartile Range (IQR) method and boxplots were used to detect outliers.

A large number of observations were identified as outliers. Removing all outliers would result in substantial information loss because some extreme values may represent actual business situations.

Instead of deleting observations, logarithmic transformation was applied because it compresses large values while preserving important information.

Log transformation reduced skewness and improved model stability.

Recommendation:

Businesses should carefully investigate large sales values because some extreme observations may indicate successful marketing campaigns or unusual customer behavior.

Feature Engineering Decision

Raw variables do not always provide sufficient information for machine learning algorithms. Therefore, feature engineering was performed to create meaningful variables from existing data.

Date-based features extracted included:

- Month
- Weekday
- Quarter
- Year

These features help identify seasonal effects and recurring purchasing patterns.

For example:

Month feature may capture holiday season sales.

Weekday may identify differences between weekday and weekend purchasing behavior.

Quarter feature may capture business cycles.

Recommendation:

Additional external variables such as holidays, festivals, weather conditions, and economic indicators should be included because they may significantly improve forecasting accuracy.

Categorical Variable Encoding Decision

Machine learning algorithms require numerical inputs; therefore categorical variables such as category, region, market, and order priority required transformation.

One-Hot Encoding was selected because it converts categorical variables into binary indicators without introducing artificial ordering.

After encoding, the number of columns increased substantially because each category generated multiple new variables.

Recommendation:

Organizations should monitor feature expansion because excessive variables increase computational cost and may introduce noise.

Feature Selection Decision

Following One-Hot Encoding, the dataset contained a very large number of features. Large feature sets increase computational requirements and may introduce unnecessary complexity.

Feature selection was performed to retain only useful and informative variables that contribute strongly to prediction performance.

Variables considered less relevant were removed.

Benefits achieved:

- Reduced model complexity
- Reduced computational cost
- Improved interpretability
- Improved training speed

Recommendation:

Feature importance analysis should be performed regularly to identify variables contributing most strongly to prediction.

Baseline Model Selection Decision

Multiple Linear Regression was selected as the baseline model because it provides a simple starting point for evaluating model performance.

Advantages:

- Easy implementation
- Fast training
- High interpretability
- Useful benchmark

However, linear regression assumes linear relationships among variables.

Recommendation:

Baseline models should always be developed before advanced algorithms because they provide useful reference points.

Advanced Model Selection Decision

Random Forest Regression was selected because sales forecasting often involves nonlinear relationships and interactions among variables.

Random Forest builds multiple decision trees and combines their outputs to improve prediction accuracy.

Advantages include:

- Captures nonlinear relationships
- Handles feature interactions
- Reduces overfitting
- Provides feature importance scores
- Performs well on large datasets

Random Forest achieved stronger performance than Linear Regression.

Recommendation:

Businesses with complex sales patterns should consider ensemble methods because they often outperform simpler models.

Hyperparameter Tuning Decision

Model performance can vary depending on parameter settings.

GridSearchCV was used for hyperparameter optimization to identify better combinations of:

- Number of estimators
- Maximum tree depth
- Minimum samples split
- Minimum samples leaf

Although tuning improved performance only slightly, results indicated that the initial Random Forest configuration was already strong.

Recommendation:

Hyperparameter tuning should be used because small improvements may become valuable at large business scales.

Future Forecasting Decision

Future input variables are generally unknown at prediction time.

Artificially generating increasing or decreasing future trends may create unrealistic forecasts.

Therefore, recent historical patterns from the dataset were used as future inputs.

This approach preserved realistic behavior and reduced forecasting bias.

Recommendation:

Businesses should continuously update models using recent data because customer behavior changes over time.

Business Recommendations

Based on forecasting results, several recommendations were developed.

Inventory Recommendation:

Maintain sufficient inventory for products with consistently high demand to avoid shortages and customer dissatisfaction.

Marketing Recommendation:

Optimize discount strategies carefully because excessive discounts may increase sales volume but reduce profitability.

Operational Recommendation:

Use forecasting results for staffing and operational planning to allocate resources efficiently.

Financial Recommendation:

Integrate forecasting outputs into budgeting and financial planning activities.

Strategic Recommendation:

Monitor highly influential product categories because they contribute significantly to revenue generation.

Technology Recommendation:

Implement automated forecasting systems for continuous prediction and monitoring.

Future Recommendation:

Explore advanced forecasting approaches including ARIMA, LSTM, and deep learning models to further improve prediction performance.